

目录

联合国教科文组织

信使

Editorial

For a long time, local knowledge has been ignored by the scientific community and sometimes even resisted. Now, it is regaining attention, and this shift is timely: the various consequences of global mechanization, along with crises related to climate change and health issues, are prompting us to re-examine those knowledge systems, practices, and expertise that have long been marginalized.

Local knowledge is rooted in detailed observations of ecosystems and is passed down through generations orally, providing a valuable key to building a sustainable future. In recent years, this type of knowledge has been increasingly recognized. The United Nations Declaration on the Rights of Indigenous Peoples, adopted in 2007, affirmed the rights of indigenous peoples to own, control, and develop their traditional knowledge.

UNESCO has long played a pioneering role in this field. Through the "Local and Indigenous Knowledge Systems" (LINKS) program, the organization actively promotes exchanges between indigenous knowledge holders and scientists, particularly in areas such as ecological protection, water resource management, and natural disaster prevention. Biosphere reserves are a concrete example of this concept—bringing together researchers and local communities to jointly explore locally adapted solutions. This effort marks a significant shift: science is no longer seen as a singular, top-down form of knowledge, but as a bridge for dialogue between different knowledge systems. Additionally, the Convention for the Safeguarding of the Intangible Cultural Heritage protects practices, techniques, and skills passed down through generations, recognizing their universal significance.

Local knowledge not only provides empirical data but also challenges the foundations of scientific paradigms from the perspectives of ethical values, the relationship between humans and nature, and long-term observational dimensions. In the fields of agriculture, traditional medicine, and climate science, such knowledge has long fostered numerous major innovations, whose sustainability often surpasses purely technology-driven solutions.

However, this recognition must be accompanied by safeguards—ensuring the community's free and fully informed consent, achieving fair distribution of benefits, and preventing exploitative plundering. Our goal is not merely to integrate local knowledge into the mainstream scientific system, but to promote cooperation that benefits all parties.

By promoting this type of knowledge, UNESCO reveals to us an unspoken truth: to understand and protect this world, we need to fully integrate the knowledge and traditions of all ethnic groups, which will bring us immeasurable benefits. True science, along with ethical standards that align with this diverse world, should provide inclusive space for the expression of all forms of human civilization, allowing the vibrant flowers of human civilization to bloom freely.

Editor-in-Chief

Agnes Bardón

2

Wide Angle

How Local Knowledge Drives Scientific Discovery

Indigenous Knowledge in the Context of Changing Times 2

Lagipoiva Cherele Jackson

Brazil: Inspiration from the Water People.7

Marcelo Silva de Sousa

When Biopiracy Breeds and Spreads.10

Daniel Robinson and David Jefferson

China: The Vibrant Revival of Traditional Dai Medicine 12

Yang Sha and Du Junzhi

The Mysterious Skies of the Kalahari.15

Sisco Auala

Sami People: Indispensable Guardians of Climate Change18

Anna Ruohonen

Ola Marek-Martinez: "Our knowledge has long been considered folklore".21

Katerina Markelova conducts the interview

24

Focus

Examining the Phenomenon of Exile from a Child's Perspective 24

Image: Darkroom of the House of
Photography Text: Katerina
Markelova

34

Viewpoint

Welcome to the Era of the Second Quantum Revolution35

Jim Al-Khalili

38

Guest

Chimamanda Ngozi Adichie: "Novels are humanity's

The final frontier of collective exploration in the realm of honest narratives" ... 39

Laetitia Kaci conducted the interview

42

Deep Reading

African book industry: Opening a new chapter42

Anuliina Savolainen

In the Context of Changing Times

Lagipoiva Cherelle Jackson

This indigenous climate journalist from Samoa has long reported on Pacific Islands issues for *The Guardian* and is currently a professor of Pacific Island Studies at Portland State University in the United States.

From indigenous peoples using controlled burns to prevent wildfires, to the Inuit's weather prediction methods, and the 'zai' (zai) water harvesting technique used in some African countries, traditional indigenous practices around the world are diverse and effective, embodying the wisdom of indigenous peoples. In the context of climate change and the ongoing loss of biodiversity, this knowledge is particularly valuable.



▼ The embroidery work of Ecuadorian photographer Carolina Zambrano, made using Chambira palm fibers—this palm is a symbol of Amazonian indigenous craftsmanship.

Indigenous Knowledge



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Brazil: Insights from the Water People

The Arapaima is crucial to the survival of the Paumari people, but its population once declined sharply. In response, the Paumari in Amazonas participated in a species recovery program that combines scientific knowledge with traditional wisdom. A few years later, the Purus River has once again seen the presence of this fish species.

On the Purus River, as twilight fades, the heat still lingers over the water's surface, and an Arapaima emerges, gulping air. This deep, moist, ancient sound awakens the nearly extinct vitality of this land. "Nature is calling for our help," recalls the young fisherman Chico Paumari, "When we rushed to the lake, we found it empty. Seeing our livelihood slipping away,

It's heartbreaking, I really don't know how to feed the children.

At dawn, the air is filled with a rusty, humid scent, intertwined with the smell of charcoal smoke. The Paumari are an indigenous group of about two thousand people, residing on the vast lands of Amazonas in northwestern Brazil. Their definition of themselves cannot be found in scientific textbooks: "We are the Water People, the only ones truly living in the water."

In 2009, the world's largest freshwater fish, the Arapaima, had only 266 adult individuals remaining, threatening the survival of the Paumari people.

For a long time, academic research and traditional knowledge have operated independently, without integration. Technicians employ scientific methods, while indigenous communities rely on intuitive perception. A turning point emerged when both sides decided to 'join forces'—this alliance was originally commemorated by Brazil's oldest institution to mark the 100th anniversary of the birth of quantum mechanics.



▼ During the fishing season, the Paumari work in shifts continuously to quickly transport the Arapaima from the lakes to the cold chain system.